



# Cambridge International AS & A Level

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## CHEMISTRY

**9701/42**

Paper 4 A Level Structured Questions

**February/March 2026**

MARK SCHEME

Maximum Mark: 100

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| <b>Published</b> |
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This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the February/March 2026 series for most Cambridge IGCSE, Cambridge International A and AS Level components, and some Cambridge O Level components.

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This document consists of **18** printed pages.

**PUBLISHED****Generic Marking Principles**

These general marking principles must be applied by all examiners when marking candidate answers. They should be applied alongside the specific content of the mark scheme or generic level descriptions for a question. Each question paper and mark scheme will also comply with these marking principles.

**GENERIC MARKING PRINCIPLE 1:**

Marks must be awarded in line with:

- the specific content of the mark scheme or the generic level descriptors for the question
- the specific skills defined in the mark scheme or in the generic level descriptors for the question
- the standard of response required by a candidate as exemplified by the standardisation scripts.

**GENERIC MARKING PRINCIPLE 2:**

Marks awarded are always **whole marks** (not half marks, or other fractions).

**GENERIC MARKING PRINCIPLE 3:**

Marks must be awarded **positively**:

- marks are awarded for correct/valid answers, as defined in the mark scheme. However, credit is given for valid answers which go beyond the scope of the syllabus and mark scheme, referring to your Team Leader as appropriate
- marks are awarded when candidates clearly demonstrate what they know and can do
- marks are not deducted for errors
- marks are not deducted for omissions
- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

**GENERIC MARKING PRINCIPLE 4:**

Rules must be applied consistently, e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

**GENERIC MARKING PRINCIPLE 5:**

Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).

**GENERIC MARKING PRINCIPLE 6:**

Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

**Science-Specific Marking Principles**

1 Examiners should consider the context and scientific use of any keywords when awarding marks. Although keywords may be present, marks should not be awarded if the keywords are used incorrectly.

2 The examiner should not choose between contradictory statements given in the same question part, and credit should not be awarded for any correct statement that is contradicted within the same question part. Wrong science that is irrelevant to the question should be ignored.

3 Although spellings do not have to be correct, spellings of syllabus terms must allow for clear and unambiguous separation from other syllabus terms with which they may be confused (e.g. ethane / ethene, glucagon / glycogen, refraction / reflection).

4 The error carried forward (ecf) principle should be applied, where appropriate. If an incorrect answer is subsequently used in a scientifically correct way, the candidate should be awarded these subsequent marking points. Further guidance will be included in the mark scheme where necessary and any exceptions to this general principle will be noted.

5 'List rule' guidance

For questions that require ***n*** responses (e.g. State **two** reasons ...):

- The response should be read as continuous prose, even when numbered answer spaces are provided.
- Any response marked *ignore* in the mark scheme should not count towards ***n***.
- Incorrect responses should not be awarded credit but will still count towards ***n***.
- Read the entire response to check for any responses that contradict those that would otherwise be credited. Credit should **not** be awarded for any responses that are contradicted within the rest of the response. Where two responses contradict one another, this should be treated as a single incorrect response.
- Non-contradictory responses after the first ***n*** responses may be ignored even if they include incorrect science.

**6** Calculation specific guidance

Correct answers to calculations should be given full credit even if there is no working or incorrect working, **unless** the question states 'show your working'.

For questions in which the number of significant figures required is not stated, credit should be awarded for correct answers when rounded by the examiner to the number of significant figures given in the mark scheme. This may not apply to measured values.

For answers given in standard form (e.g.  $a \times 10^n$ ) in which the convention of restricting the value of the coefficient ( $a$ ) to a value between 1 and 10 is not followed, credit may still be awarded if the answer can be converted to the answer given in the mark scheme.

Unless a separate mark is given for a unit, a missing or incorrect unit will normally mean that the final calculation mark is not awarded. Exceptions to this general principle will be noted in the mark scheme.

**7** Guidance for chemical equations

Multiples / fractions of coefficients used in chemical equations are acceptable unless stated otherwise in the mark scheme.

State symbols given in an equation should be ignored unless asked for in the question or stated otherwise in the mark scheme.

**Annotations guidance for centres**

Examiners use a system of annotations as a shorthand for communicating their marking decisions to one another. Examiners are trained during the standardisation process on how and when to use annotations. The purpose of annotations is to inform the standardisation and monitoring processes and guide the supervising examiners when they are checking the work of examiners within their team. The meaning of annotations and how they are used is specific to each component and is understood by all examiners who mark the component.

We publish annotations in our mark schemes to help centres understand the annotations they may see on copies of scripts. Note that there may not be a direct correlation between the number of annotations on a script and the mark awarded. Similarly, the use of an annotation may not be an indication of the quality of the response.

The annotations listed below were available to examiners marking this component in this series.

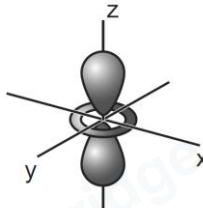
**Annotations**

| <b>Annotation</b>                                                                   | <b>Meaning</b>                                                                 |
|-------------------------------------------------------------------------------------|--------------------------------------------------------------------------------|
|    | Correct point or mark awarded                                                  |
|    | Incorrect point or mark not awarded                                            |
|    | Unclear                                                                        |
|    | Information missing or insufficient for credit                                 |
|  | Benefit of the doubt given                                                     |
|  | Contradiction in response otherwise markworthy, mark not given                 |
|  | Part of the correct answer has been seen. Full credit has not been awarded.    |
|  | Error carried forward applied                                                  |
|  | Incorrect or insufficient point ignored while marking the rest of the response |
|  | Rounding error                                                                 |

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| Annotation  | Meaning                                |
|-------------|----------------------------------------|
| <b>REP</b>  | Repetition                             |
| <b>SEEN</b> | Blank page or part of script seen      |
| <b>SF</b>   | Error in number of significant figures |
| <b>TE</b>   | Transcription error                    |

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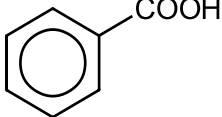
| Question  | Answer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Marks |
|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 1(a)(i)   | (d-block element which forms one or more) stable ions/compounds with an incomplete d orbital(s)/d sub-shell                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 1     |
| 1(a)(ii)  | <div style="display: flex; justify-content: space-around; align-items: center;"> <div style="text-align: center;"> <b>3d</b><br/> <div style="display: flex; gap: 5px;"> <div style="border: 1px solid black; padding: 2px 10px;">1</div> <div style="border: 1px solid black; padding: 2px 10px;">1</div> <div style="border: 1px solid black; padding: 2px 10px;">1</div> <div style="border: 1px solid black; padding: 2px 10px;">1</div> <div style="border: 1px solid black; padding: 2px 10px;">1</div> </div> </div> <div style="text-align: center;"> <b>4s</b><br/> <div style="border: 1px solid black; padding: 2px 10px;">1</div> </div> <div style="text-align: center;"> <b>4p</b><br/> <div style="display: flex; gap: 5px;"> <div style="border: 1px solid black; width: 30px; height: 20px;"></div> <div style="border: 1px solid black; width: 30px; height: 20px;"></div> <div style="border: 1px solid black; width: 30px; height: 20px;"></div> </div> </div> </div> | 1     |
| 1(a)(iii) |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 1     |
| 1(a)(iv)  | <p>more than one (stable) oxidation state / exist in variable oxidation states</p> <p>vacant/empty (d) orbitals are energetically accessible</p> <p><b>OR</b></p> <p>vacant/empty (d) orbitals can form dative bonds with ligands</p> <p><b>OR</b></p> <p>vacant/empty (d) orbitals can accept lone pair to bond with ligands</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 2     |
| 1(b)(i)   | <p><b>M1:</b> emf / potential / voltage of an electrode / half-cell compared / connected to SHE / hydrogen half-cell</p> <p><b>M2:</b> 1 mol dm<sup>-3</sup>, 1 atm, 298 K</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 2     |

| Question  | Answer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Marks |
|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 1(b)(ii)  | <p><b>M1:</b> (<math>\text{Cr}^{2+}</math> ions) can be oxidised to <math>\text{Cr}^{3+}</math></p> <p><b>M2:</b> by <math>\text{O}_2</math> in acidic conditions / <math>\text{O}_2</math> with <math>\text{H}^+</math> because <math>E^\ominus_{\text{cell}} = (+)1.64 \text{ (V)}</math></p> <p><b>OR</b></p> <p>by <math>\text{O}_2</math> in acidic conditions / <math>\text{O}_2</math> with <math>\text{H}^+</math> as <math>E^\ominus \text{O}_2 / \text{H}^+</math> is more positive than <math>\text{Cr}^{3+} / \text{Cr}^{2+}</math></p> <p><b>OR</b></p> <p>by <math>\text{O}_2</math> in alkaline conditions / <math>\text{O}_2</math> with <math>\text{OH}^-</math> because <math>E^\ominus_{\text{cell}} = (+) 1.66 \text{ (V)}</math> <b>OR</b> <math>+0.81 \text{ V}</math></p> <p><b>OR</b></p> <p>by <math>\text{O}_2</math> in alkaline conditions / <math>\text{O}_2</math> with <math>\text{OH}^-</math> as <math>E^\ominus \text{O}_2 / \text{OH}^-</math> is more positive than <math>\text{Cr}^{3+} / \text{Cr}^{2+}</math></p> | 2     |
| 1(b)(iii) | <p><i>for green solid</i></p> <p><math>\text{Cr}^{3+}(\text{aq}) + 3\text{OH}^-(\text{aq}) \rightarrow \text{Cr}(\text{OH})_3(\text{s})</math></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 1     |
| 1(b)(iv)  | <p><i>for yellow solution</i></p> <p><math>4\text{Cr}(\text{OH})_3(\text{s}) + 8\text{OH}^-(\text{aq}) + 3\text{O}_2(\text{g}) \rightarrow 4\text{CrO}_4^{2-}(\text{aq}) + 10\text{H}_2\text{O}(\text{l})</math></p> <p>correct species [1]</p> <p>correct balancing [1]</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 2     |
| 1(c)(i)   | <p>gaseous products are formed</p> <p><b>OR</b></p> <p>increase in moles / molecules of gas</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 1     |
| 1(c)(ii)  | <p><math>915 = 192 + 81 + 4S - 114</math></p> <p><math>\therefore S = \frac{1}{4} \times 756 = (+) 189 \text{ (J K}^{-1} \text{ mol}^{-1}\text{)}</math></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 2     |
| 1(c)(iii) | <p><math>\Delta G</math> always / needs / is to be negative / <math>&lt; 0</math> because <math>\Delta H &lt; 0</math> / negative / exothermic <math>\Delta S &gt; 0</math> / positive / <math>-\text{T}\Delta S</math> is negative</p> <p>high activation energy / <math>E_a</math> prevents reaction</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 2     |
| 1(d)(i)   | <p>(enthalpy change/energy released when)</p> <ul style="list-style-type: none"> <li>1 mole of</li> <li>gaseous atoms accept</li> <li>one mole of <math>\text{e}^-</math></li> </ul> <p>any two [1], all three [2]</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 2     |

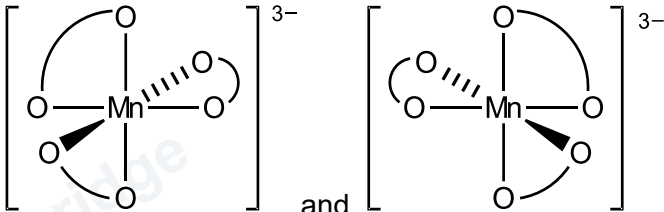


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| Question  | Answer                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Marks    |
|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 1(d)(ii)  | <b>M1:</b> $(2 \times 398 + 3 \times 248 + 2 \times 5233 + 3 \times 702)$<br><b>M2:</b> $-1128 - 14112 = -15240 \text{ (kJ mol}^{-1}\text{) min 4 sf}$                                                                                                                                                                                                                                                                                                          | <b>2</b> |
| 1(d)(iii) | <ul style="list-style-type: none"> <li>• <math>\text{Cr}_2\text{O}_3</math> less exothermic / less negative</li> <li>• <math>\text{Cr}^{3+}</math> (has same charge but) larger radius / is less charge dense</li> <li>• less attraction between <math>\text{Cr}^{3+}</math> and <math>\text{O}^{2-}</math> / the different ions <b>OR</b> weaker bond between <math>\text{Cr}^{3+}</math> and <math>\text{O}^{2-}</math></li> </ul> any two [1], all three [2] | <b>2</b> |

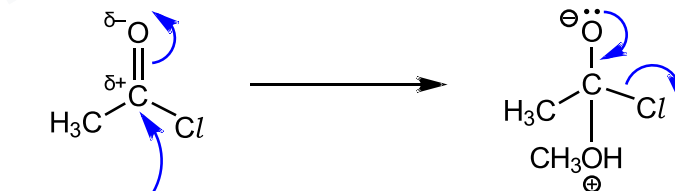
| Question | Answer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Marks |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 2(a)(i)  | <p><b>M1:</b> moles of manganate = <math>23.80 / 1000 \times 0.0500</math> <b>OR</b> <math>1.19 \times 10^{-3}</math> mol<br/> <b>OR</b> initial moles of <math>\text{Fe}^{2+}</math> = <math>17.70 \div 277.9</math> <b>OR</b> <math>6.37 \times 10^{-2}</math> mol</p> <p><b>M2:</b> moles of <math>\text{Fe}^{2+}</math> in aliquot = <math>5 \times 1.19 \times 10^{-3}</math> <b>OR</b> <math>5.95 \times 10^{-3}</math> mol<br/> <b>AND</b><br/> total moles of <math>\text{Fe}^{2+}</math> = <math>5.95 \times 10^{-3} \times 10</math> <b>OR</b> <math>5.95 \times 10^{-2}</math> mol<br/> <b>OR</b><br/> mass of <math>\text{FeSO}_4 \cdot 7\text{H}_2\text{O}</math> left = <math>5.95 \times 10^{-3} \times 277.9</math> <b>OR</b> 1.6535 g (in 25 cm<sup>3</sup>)</p> <p><b>M3:</b> mass of <math>\text{FeSO}_4 \cdot 7\text{H}_2\text{O}</math> left = <math>1.6535 \text{ g} \times 10</math> = 16.535 g (in 250 cm<sup>3</sup>)<br/> <b>OR</b><br/> mass of <math>\text{FeSO}_4 \cdot 7\text{H}_2\text{O}</math> left = <math>277.9 \times 5.95 \times 10^{-2}</math> = 16.535 g</p> <p>percentage oxidised = <math>100\% \times (17.70 - 16.535) / 17.70</math><br/> = <math>100\% \times 1.165 / 17.70</math> = 6.58 % min 2sf</p> | 3     |
| 2(a)(ii) | <p><math>\text{MnO}_4^-</math> / manganate is a (self-)indicator<br/> <b>OR</b><br/> <math>\text{MnO}_4^-</math> / manganate changes colour / purple to colourless / decolourises<br/> <b>OR</b><br/> solution changes (from colourless) to pink / purple at the end-point</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 1     |
| 2(b)(i)  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 1     |
| 2(c)     | <p>manganese is both oxidised and reduced (to a single oxidation state)</p> <p>from +VII / +7 to +III / +3 <b>AND</b> +II / +2 to +III / +3</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 2     |
| 2(d)(i)  | $K_{\text{sp}} = [\text{Mn}^{2+}] [\text{OH}^-]^2$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 1     |

| Question | Answer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Marks |
|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 2(d)(ii) | $[\text{Mn}^{2+}(\text{aq})] = 3.20 \times 10^{-3} / 88.9$ <b>OR</b> $3.60 \times 10^{-5} \text{ mol dm}^{-3}$<br>$K_{\text{sp}} = 3.60 \times 10^{-5} \times (2 \times 3.60 \times 10^{-5})^2 = 4 \times (3.60 \times 10^{-5})^3$<br>$= 1.87 \times 10^{-13} \text{ min 2sf}$<br>$\text{mol}^3 \text{ dm}^{-9}$                                                                                                                                                                                                                                            | 3     |
| 2(e)(i)  | <b>M1:</b> d orbital(s) of two different energies / d-d splitting occurs<br><b>OR</b><br>d orbital(s)/d (sub)-shell splits / d-d gap<br><br><b>M2:</b> electron(s) promoted / excited<br><b>OR</b><br>electron(s) moves to higher (d–) orbital<br><br><b>M3:</b> wavelength / frequency / light / photon / $h\nu$ / $hf$ absorbed<br><b>OR</b><br>radiation / energy from <u>visible</u> (region) absorbed <b>AND</b> colour (seen) is complementary<br><b>OR</b><br>wavelength / frequency / colour / light not absorbed is transmitted / reflected / seen | 3     |

| Question | Answer                  |                                                                                    | Marks |
|----------|-------------------------|------------------------------------------------------------------------------------|-------|
| 2(e)(ii) | formula                 | $[\text{Mn}(\text{C}_2\text{O}_4)_3]^{3-}$                                         | 4     |
|          | formula of ligand       | $\text{C}_2\text{O}_4^{2-}$                                                        |       |
|          | shape                   | octahedral                                                                         |       |
|          | coordination number     | 6                                                                                  |       |
|          | type of stereoisomerism | optical                                                                            |       |
|          | structures              |  |       |

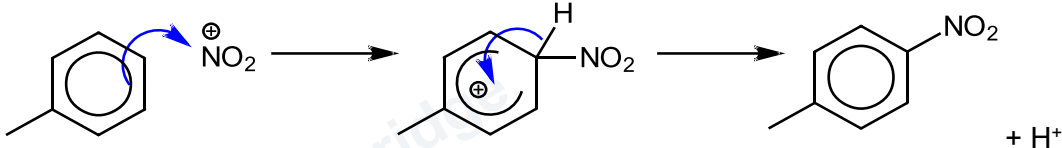
| Question | Answer                                                                                                                                                                                                                    | Marks |
|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 3(a)     |                                                                                                                                                                                                                           | 2     |
| 3(b)     | $\text{HO}(\text{CH}_2)_3\text{COOH} + 2\text{PCl}_5 \rightarrow \text{Cl}(\text{CH}_2)_3\text{COCl} + 2\text{POCl}_3 + 2\text{HCl}$<br>organic product $\text{Cl}(\text{CH}_2)_3\text{COCl}$ [1]<br>rest of equation [1] | 2     |

| Question  | Answer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Marks    |
|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 3(c)(i)   | $[H^+] = (K_a \times [HA])^{1/2} = (1.91 \times 10^{-5} \times 0.150)^{1/2}$ <b>OR</b> $1.69 \times 10^{-3} \text{ (mol dm}^{-3}\text{)}$<br>$\text{pH} = -\log [H^+] = 2.77 \text{ min 2sf}$                                                                                                                                                                                                                                                                                                                                                                                                                             | <b>2</b> |
| 3(c)(ii)  | (a solution that) resists / opposes / minimises changes in pH<br>when small (amounts / volumes) of acid / $H^+$ and base / alkali / $OH^-$ are added to it                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>2</b> |
| 3(c)(iii) | $HO(CH_2)_3COO^-$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <b>1</b> |
| 3(c)(iv)  | <b>M1:</b> moles of HA initially $= (0.150 \times 25.0 / 1000)$ <b>OR</b> $3.75 \times 10^{-3} \text{ mol}$<br><b>AND</b><br>moles of NaOH / moles of $A^-$ $= (0.200 \times 10.0 / 1000)$ <b>OR</b> $2.00 \times 10^{-3} \text{ mol}$<br><b>M2:</b> remaining moles of HA $= 3.75 \times 10^{-3} - 2.00 \times 10^{-3}$ <b>OR</b> $1.75 \times 10^{-3} \text{ mol}$<br><b>M3:</b> $[H^+] = K_a[HA]/[A^-] = 1.91 \times 10^{-5} \times \frac{(1.75 \times 10^{-3} \div 0.035)}{(2.00 \times 10^{-3} \div 0.035)} = 1.67 \times 10^{-5}$<br><b>AND</b> $\text{pH} = -\log_{10} 1.67 \times 10^{-5} = 4.78 \text{ min 2sf}$ | <b>3</b> |
| 3(d)      | (an excess of) acidified $K_2Cr_2O_7$ / $KMnO_4$ <b>AND</b> heat / boil / hot / reflux                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <b>1</b> |
| 3(e)(i)   | (the) ratio of the concentrations of a solute between two solvents at equilibrium                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <b>1</b> |
| 3(e)(ii)  | <b>M1:</b> $[\text{acid(ether)}] = 0.0375 \text{ mol dm}^{-3}$ <b>AND</b> $[\text{acid(aq)}] = 0.2775 \text{ mol dm}^{-3}$<br><b>M2:</b> $K_{pc} = [\text{acid(ether)}] / [\text{acid(aq)}] = 0.135$                                                                                                                                                                                                                                                                                                                                                                                                                      | <b>2</b> |

| Question  | Answer                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Marks |
|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 4(a)(i)   | two separate half-life readings quoted $t_{1/2} \cong 6000 \text{ (s)} \pm 250$<br>same / constant half-life ( $\therefore$ first order with respect to [ester])                                                                                                                                                                                                                                                                                                            | 2     |
| 4(a)(ii)  | tangent seen at $t = 0$ <b>AND</b> initial rate (from gradient of tangent) $\cong 0.24 / 8750 = 2.74 \times 10^{-5} \text{ (mol dm}^{-3} \text{ s}^{-1})$ min 2sf                                                                                                                                                                                                                                                                                                           | 1     |
| 4(a)(iii) | rate = $k [\text{CH}_3\text{COOCH}_3] [\text{H}^+]$<br>$\text{mol}^{-1} \text{ dm}^3 \text{ s}^{-1}$                                                                                                                                                                                                                                                                                                                                                                        | 2     |
| 4(b)(i)   | <ul style="list-style-type: none"> <li>reaction 6 / <math>\text{CH}_3\text{COCl}</math> goes to completion / non-reversible</li> <li>reaction 7 / <math>\text{CH}_3\text{COOH}</math> is an equilibrium (reaction) / reversible</li> <li>reaction 6 has a gas formed <b>AND</b> has a greater <math>\Delta S</math> / entropy is increased (so more feasible)</li> </ul> any two [2]                                                                                        | 2     |
| 4(b)(ii)  |  <p>CH<sub>3</sub>OH</p> <p><b>M1:</b> correct arrow from lone pair on O (in CH<sub>3</sub>OH) to C (of C=O)</p> <p><b>M2:</b> correct dipole on C=O <b>AND</b> correct arrow from the C=O bond to O atom</p> <p><b>M3:</b> correct intermediate (both charges present)</p> <p><b>M4:</b> arrow from (lone pair on) O<sup>(-)</sup> to C–O bond <b>AND</b> arrow from C–Cl bond to Cl</p> | 4     |

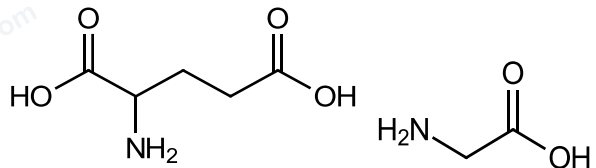
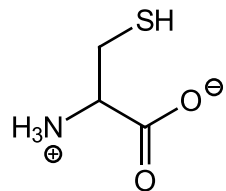
| Question | Answer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Marks |
|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 4(c)     | <p><b>M1:</b> chlorobenzene &lt; chloroethane &lt; ethanoyl chloride</p> <p><b>M2 &amp; M3:</b> Any two from:<br/> <i>(acyl chlorides easier than alkyl chlorides)</i></p> <ul style="list-style-type: none"> <li>has two electronegative atom groups/an extra electronegative atom / an electronegative oxygen / electron withdrawing C=O <b>AND</b> so C (of C=O) highly / more electron deficient / C-Cl bond weak(est)</li> </ul> <p><i>(aryl chlorides hardest)</i></p> <ul style="list-style-type: none"> <li>LP / lone pair / p-orbital on Cl/chlorine <b>AND</b> overlaps / delocalised with ring / <math>\pi</math> system <b>AND</b> (C-Cl) bond is strengthened / has partial double bond character</li> </ul> <p><i>(alkyl chlorides easier than aryl chlorides)</i></p> <ul style="list-style-type: none"> <li>carbon is bonded to an electronegative atom / Cl / not bonded to electronegative O <b>AND</b> so has less electron deficient carbon</li> </ul> <p><b>OR</b></p> <ul style="list-style-type: none"> <li>C-Cl bond strengthened <b>AND</b> by positive inductive effect of alkyl / R group</li> </ul> | 3     |

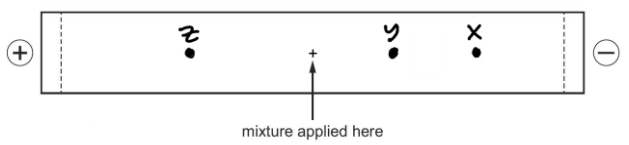
| Question | Answer                                                                                                                                                                                                                                | Marks |
|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 5(a)(i)  | $\text{HNO}_3 + \text{H}_2\text{SO}_4 \rightarrow {}^+\text{NO}_2 + \text{H}_2\text{O} + \text{HSO}_4^-$<br><b>OR</b><br>$\text{HNO}_3 + 2\text{H}_2\text{SO}_4 \rightarrow {}^+\text{NO}_2 + \text{H}_3\text{O}^+ + 2\text{HSO}_4^-$ | 1     |

| Question  | Answer                                                                                                                                                                                                                                                                                                                                                | Marks |
|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 5(a)(ii)  |  <p><b>M1:</b> curly arrow 1 from touching / inside the hexagon to <u>nitrogen</u> of NO<sub>2</sub><sup>+</sup></p> <p><b>M2:</b> correct intermediate</p> <p><b>M3:</b> curly arrow 2 from C-H bond into hexagon (touching / inside) <b>AND</b> H<sup>+</sup></p> | 3     |
| 5(a)(iii) | CH <sub>3</sub> C <sub>6</sub> H <sub>4</sub> NO <sub>2</sub> + 6[H] → CH <sub>3</sub> C <sub>6</sub> H <sub>4</sub> NH <sub>2</sub> + 2H <sub>2</sub> O                                                                                                                                                                                              | 1     |
| 5(a)(iv)  | Stage 1 <b>M1:</b> HNO <sub>2</sub> <b>OR</b> NaNO <sub>2</sub> with (dilute strong) acid <b>AND</b> ≤ 10 °C<br>Stage 2 <b>M2:</b> warming with H <sub>2</sub> O / aqueous                                                                                                                                                                            | 2     |
| 5(a)(v)   | CH <sub>3</sub> Cl <b>AND</b> AlCl <sub>3</sub><br>NO <sub>2</sub> directs to the meta / 3(,5)-position (in the ring so 4-methylation unlikely)                                                                                                                                                                                                       | 2     |
| 5(b)(i)   | it is inert / unreactive                                                                                                                                                                                                                                                                                                                              | 1     |
| 5(b)(ii)  | a high-boiling point / non-volatile <b>AND</b> (non-polar) liquid / solvent                                                                                                                                                                                                                                                                           | 1     |
| 5(b)(iii) | it is the least polar<br><b>OR</b><br>it is more attracted to the stationary phase<br><b>OR</b><br>forms a strong IMF to the stationary phase<br><b>OR</b><br>attracted to the stationary phase for a longer time<br><b>OR</b><br>it is the least volatile / highest bp                                                                               | 1     |



| Question | Answer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Marks |
|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 5(b)(iv) | $\% = 100\% \times 22 / (29 + 34 + 22) = 25.88$<br>$\% = 25.9$ <b>OR</b> 26 min 2sf                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 1     |
| 5(c)     | <p><b>M1:</b> weakest = ethanol ... water ... phenol = strongest</p> <p><b>M2:</b> <i>why acids are stronger or weaker</i><br/>           strongest acid has weakest O—H<br/> <b>OR</b><br/>           H<sup>+</sup> more easily lost<br/> <b>OR</b><br/>           anion stabilised / conjugate base stabilised</p> <p><b>M3:</b> <i>for phenol:</i><br/>           as p-orbital / lone pair on oxygen overlaps / delocalised into the ring / <math>\pi</math> system / delocalised system</p> <p><b>M4:</b> <i>for ethanol:</i><br/>           positive inductive effect/electron donating of alkyl / R group</p> | 4     |

| Question  | Answer                                                                                     | Marks |
|-----------|--------------------------------------------------------------------------------------------|-------|
| 6(a)(i)   |          | 2     |
| 6(a)(ii)  | pH at which a molecule is neutral / has no overall charge / charge of zero / no net charge | 1     |
| 6(a)(iii) |         | 1     |

| Question  | Answer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Marks |
|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 6(b)      | <p><b>M1:</b></p>  <p><b>M2:</b> direction in terms of charge of species<br/> <b>Z</b> moves towards positive (pole) <b>AND</b> is negatively charged<br/> <b>OR</b><br/> <b>X / Y</b> move towards negative (pole) <b>AND</b> is positively charged</p> <p><b>M3:</b> distance in terms of size<br/> <b>X</b> moves farther / faster (than <b>Y</b>) because of lower <math>M_r</math> / smaller molecular size</p> | 3     |
| 6(c)(i)   | <p><b>M1:</b> species with / has two lone pairs / LP (of electrons)</p> <p><b>M2:</b> that form dative (covalent) / co-ordinate bond(s) to a (central) transition-element / metal <b>AND</b> atom / ion<br/> <b>OR</b><br/> donates electron pair(s) to a (central) transition-element / metal <b>AND</b> atom / ion</p>                                                                                                                                                                               | 2     |
| 6(c)(ii)  |                                                                                                                                                                                                                                                                                                                                                                                                                     | 1     |
| 6(c)(iii) | <p>better / improved biological activity<br/> <b>OR</b><br/> reduction of dosage<br/> <b>OR</b><br/> the other isomer may have other / different / unwanted biological activity<br/> <b>OR</b><br/> the other isomer may have different / unwanted side effects / be toxic / be harmful</p>                                                                                                                                                                                                            | 1     |
| 6(c)(iv)  | $[\text{Cu}(\text{H}_2\text{O})_6]^{2+} + 2\text{Z}^- \rightarrow [\text{Cu}(\text{Z})_2(\text{H}_2\text{O})_2] + 4\text{H}_2\text{O}$                                                                                                                                                                                                                                                                                                                                                                 | 1     |